Math Training — M1 Cognitive Science

Session 1 — Linear Algebra I

Vectors, span, dot product, change of basis, matrices

Advanced track

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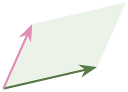
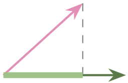
Roadmap of the session

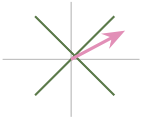
1. Vector, linear combination, **span**
2. Dot product $\longrightarrow$ **projection**
3. **Change of basis**: same point, new axes
4. **Matrix** $=$ transformation of space



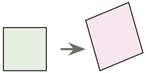
span



projection

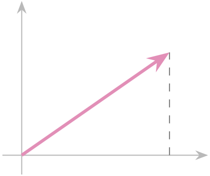


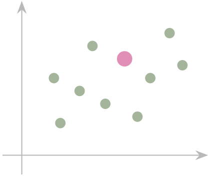
new axes



transformation

One object, three readings





$$\mathbf{p} = \begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}$$ one number per measure

arrow: length, angle

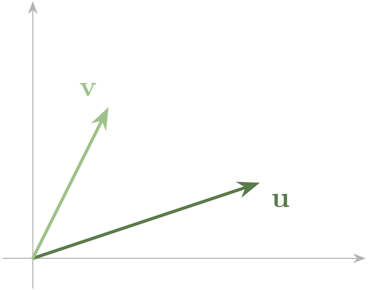
point: one row of a table

list: how data arrives

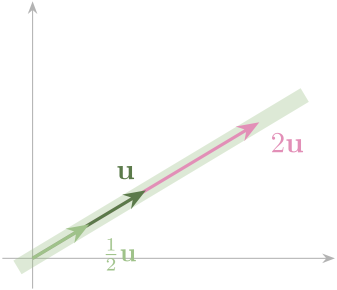
Definition — vector

$\mathbf{v} \in \mathbb{R}^n$: an ordered list of n numbers. Two operations only — **add** slot by slot, **scale** $\lambda\mathbf{v} = (\lambda v_1, \ldots, \lambda v_n)$.

Addition and scaling, seen geometrically



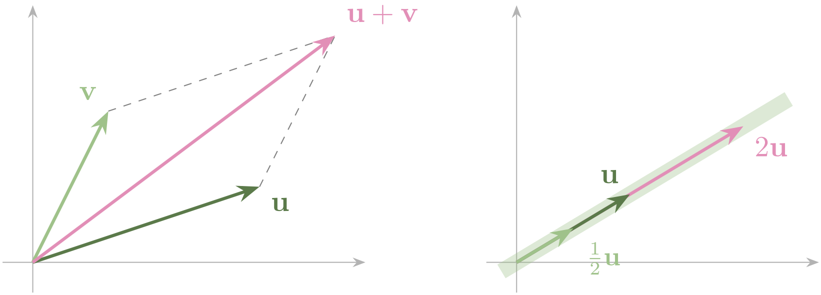
add: tip to tail



scale: never leaves the line

Left: the sum is the far corner of the parallelogram. Right: scaling slides the arrow along its own line — which is why a single *nonzero* vector spans a line.

Addition and scaling, seen geometrically



add: tip to tail

scale: never leaves the line

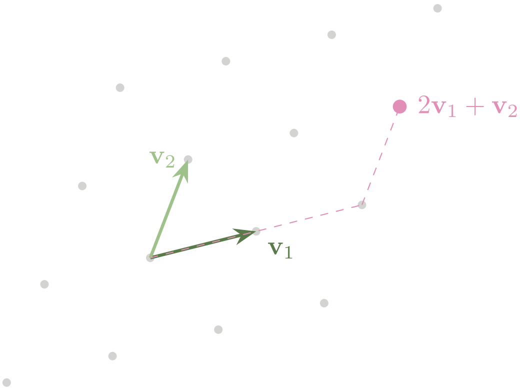
Left: the sum is the far corner of the parallelogram. Right: scaling slides the arrow along its own line — which is why a single *nonzero* vector spans a line.

Scale each one, then add them up

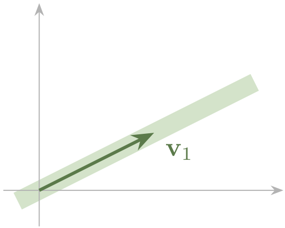
$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \cdots + \lambda_k \mathbf{v}_k.$$

All of it is this one operation:

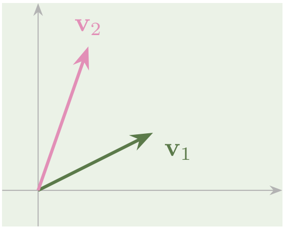
- span, dependence
- basis, coordinates
- matrix–vector product
- rank, eigenvectors
- principal components



Span: the set of all linear combinations



one vector ⇒ a **line**

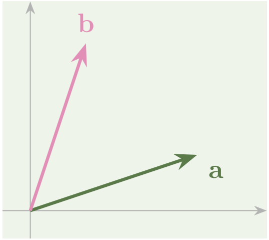


two independent ⇒ a **plane**

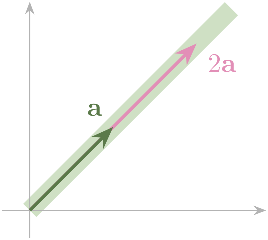
- **Definition — span**

  $\mathrm{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ is the set of *all* their linear combinations: a point, a line, a plane, or a flat of higher dimension — always containing the origin.

Linear independence and dependence



independent: the whole plane



dependent: one line only

Definition — linear dependence

Dependent $\iff$ $\lambda_1 \mathbf{v}_1 + \cdots + \lambda_k \mathbf{v}_k = \mathbf{0}$ for some λ_i *not all zero*. Independent otherwise.

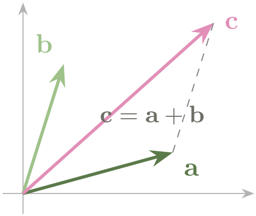
In $\mathbb{R}^n$, any $n+1$ vectors are dependent

- **Useful fact**

  In $\mathbb{R}^n$: any $n+1$ vectors are dependent, and any n independent vectors span all of $\mathbb{R}^n$.

Three vectors in $\mathbb{R}^2$? Dependent.
No computation needed.

This is the fact *rank* will rest on next session.



Pitfall: checking pairs does not test independence

- **Watch out**

  "No one is a multiple of another, so they are independent."

$$\mathbf{v}_1 = (1,2,3), \qquad \mathbf{v}_2 = (2,1,0), \qquad \mathbf{v}_3 = (4,5,6)$$

No pair is proportional — and yet $2\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3 = \mathbf{0}$. **Dependent.**

And the mirror mistake: $(1,2)$ and $(3,5)$ *look* nearly parallel, yet $\lambda_1(1,2) + \lambda_2(3,5) = \mathbf{0}$ forces $\lambda_1 = \lambda_2 = 0$: **independent**.

The cure: from three vectors on, "multiple of" is useless. Solve for a vanishing combination.

The running example: a data table

	RT	acc.	anx.
P_1	4	9	1
P_2	6	7	3
P_3	5	8	2
P_4	9	4	6
P_5	7	6	4

$\longleftarrow$ a **row** = one participant

$\uparrow$ a **column** = one measure, across everybody

One table, two vector spaces

The table is 5×3, so a row and a column cannot even live in the same space: $P_2 = (6, 7, 3) \in \mathbb{R}^3$ (one coordinate per measure), the accuracy column $(9, 7, 8, 4, 6) \in \mathbb{R}^5$ (one per participant). Which of the two you are working in is decided by the question you ask.

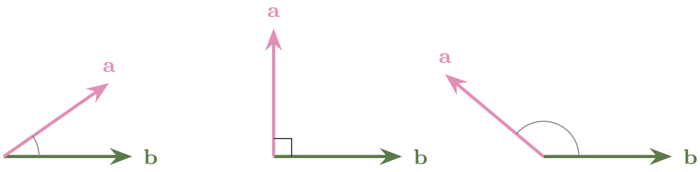
Practice — try it

1. $\mathbf{a} = (1, 2)$, $\mathbf{b} = (3, -1)$: compute $\mathbf{a} + \mathbf{b}$, $2\mathbf{a}$, $\mathbf{a} - \mathbf{b}$.
2. Independent? (i) $(1, 1)$ & $(2, 2)$; (ii) $(1, 0)$ & $(0, 1)$; (iii) $(2, -4)$ & $(-1, 2)$; (iv) $(1, 1, 0)$, $(0, 1, 1)$ & $(1, 3, 2)$.
   Give a vanishing combination for each dependent family.
3. Describe $\operatorname{span}\{(1, 1, 0), (0, 1, 1)\}$ in $\mathbb{R}^3$. Is $(1, 2, 1)$ in it?

Definition — dot product

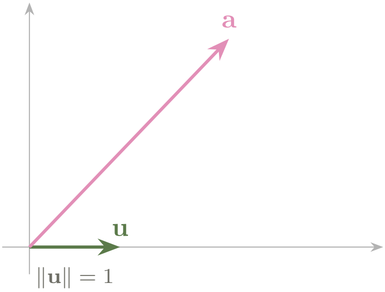
$$\mathbf{a}\cdot\mathbf{b}=\sum_{i=1}^{n} a_i b_i \;=\; \|\mathbf{a}\|\,\|\mathbf{b}\|\cos\alpha, \qquad \|\mathbf{a}\|=\sqrt{\mathbf{a}\cdot\mathbf{a}}.$$

The first expression is the definition; the second is a theorem — it is what makes the dot product geometric.



$\mathbf{a}\cdot\mathbf{b}>0$ $\quad$ $\mathbf{a}\cdot\mathbf{b}=0$ $\quad$ $\mathbf{a}\cdot\mathbf{b}<0$

The dot product as a projection

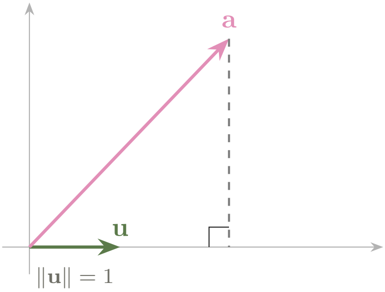


Useful fact

$$\mathbf{a}\cdot\mathbf{b} = \underbrace{\left(\|\mathbf{a}\|\cos\alpha\right)}_{\text{signed length of the shadow}} \times \|\mathbf{b}\|,$$ and for *unit* $\mathbf{u}$: $\mathbf{a}\cdot\mathbf{u} =$ signed shadow length.

A coordinate in an orthonormal basis *is* a dot product.

The dot product as a projection



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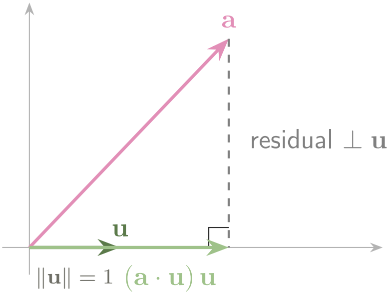

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A coordinate in an orthonormal basis *is* a dot product.

Definition — orthogonal projection

Onto the line spanned by any $\mathbf{v} \neq \mathbf{0}$, and onto a subspace S with *orthonormal* basis $\mathbf{e}_1, \ldots, \mathbf{e}_k$:

$$P_{\mathbf{v}}(\mathbf{a}) = \frac{\mathbf{a} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\, \mathbf{v} \quad \Big(\text{unit } \mathbf{u}:\ (\mathbf{a} \cdot \mathbf{u})\, \mathbf{u}\Big), \qquad P_S(\mathbf{a}) = \sum_{i=1}^{k} (\mathbf{a} \cdot \mathbf{e}_i)\, \mathbf{e}_i.$$

Useful fact — the three properties that characterise it

1. $P_S(\mathbf{a}) \in S$, and the residual $\mathbf{a} - P_S(\mathbf{a})$ is orthogonal to *every* vector of S;
2. $P_S(\mathbf{a})$ is the point of S **closest** to $\mathbf{a}$ — this is least squares;
3. $P_S\big(P_S(\mathbf{a})\big) = P_S(\mathbf{a})$: projecting twice changes nothing.

Cauchy–Schwarz, the cosine, and one caution

Cauchy—Schwarz

$|\mathbf{a}\cdot\mathbf{b}| \le \|\mathbf{a}\|\,\|\mathbf{b}\|$, with equality exactly when $\mathbf{a}$ and $\mathbf{b}$ are linearly dependent.

Divide through: $\cos\alpha \in [-1,1]$ — the *cosine similarity*.

Watch out

The dot product is **not** bounded by 1: $(10,0)\cdot(10,0) = 100$.

The bounded quantity is $\dfrac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|}$.

A weighted score: size and alignment

Example

Composite score with weights $\mathbf{w} = (1,\ 2,\ -1)$ (reward speed a little, accuracy a lot, penalise anxiety): score$(\mathbf{p}) = \mathbf{w} \cdot \mathbf{p}$.

	$\mathbf{w} \cdot \mathbf{p}$	$\lVert \mathbf{p} \rVert$	$\cos\alpha$
$\mathbf{p} = (1,1,0)$	3	$\sqrt{2}$	0.87
$\mathbf{r} = (2,2,0)$	6	$2\sqrt{2}$	0.87
$\mathbf{q} = (0,1,2)$	0	$\sqrt{5}$	0

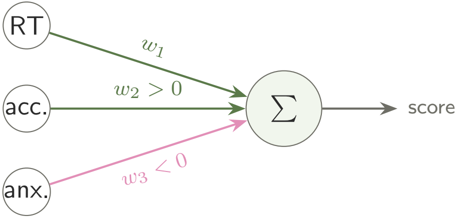
$\mathbf{r}$ scores twice $\mathbf{p}$ *only* because it is twice as long: same shape, same $\cos\alpha$.

The score is size $\times$ alignment

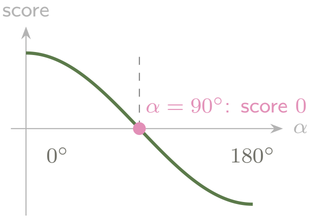
$$\underbrace{\mathbf{w} \cdot \mathbf{p}}_{\text{the score}} = \lVert \mathbf{w} \rVert \times \underbrace{\lVert \mathbf{p} \rVert \cos\alpha}_{\text{shadow of } \mathbf{p} \text{ on } \mathbf{w}}$$

The shadow $\mathbf{p} \cdot \frac{\mathbf{w}}{\lVert \mathbf{w} \rVert} = \frac{3}{\sqrt{6}} \approx 1.22$ is the same score read with a **unit ruler**: doubling all the weights doubles every score and changes nothing, and the shadow is what survives that rescaling.

The blind directions of a weighted score



three numbers in, one number out



$\mathbf{p} = (1, 1, 0)$ and $\mathbf{p}' = (3, 0, 0)$
both score 3: $\mathbf{w} \cdot (\mathbf{p}' - \mathbf{p}) = 0$

- **Two profiles are indistinguishable when their difference is $\perp \mathbf{w}$**

  Here $\mathbf{p}' - \mathbf{p} = (2, -1, 0)$ and $\mathbf{w} \cdot (\mathbf{p}' - \mathbf{p}) = 0$, so $\mathbf{p} = (1, 1, 0)$ and $\mathbf{p}' = (3, 0, 0)$ get **exactly the same score**, however different they are: the profiles with a given score fill a whole plane perpendicular to $\mathbf{w}$. Restricted to profiles of the same *size*, the score follows $\cos\alpha$ — maximal along $\mathbf{w}$, zero at $90°$. The rule is not weakly sensitive to those directions, it is blind to them.

Practice — try it

1. Compute $(1, 2, 3) \cdot (4, 0, -1)$ and $\|(3, 4)\|$.
2. Are $(1, 2)$ and $(2, -1)$ orthogonal? What does that mean for a scoring rule with weights $(1, 2)$ applied to the profile $(2, -1)$?
3. Project $\mathbf{a} = (3, 1)$ onto $\mathbf{u} = \frac{1}{\sqrt{2}}(1, 1)$: signed shadow length, then the projection vector.
4. Project $\mathbf{a} = (3, 1)$ onto $\mathbf{v} = (1, 1)$ — *not* a unit vector. Same answer?

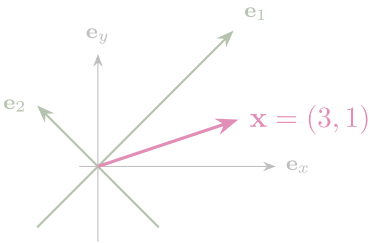
Practice — solutions

Answers

1. $(1,2,3)\cdot(4,0,-1) = 4+0-3 = 1$; $\|(3,4)\| = \sqrt{9+16} = 5$.
2. $(1,2)\cdot(2,-1) = 2-2 = 0$: orthogonal. That profile scores **zero** — the rule is completely blind to it.
3. $\mathbf{a}\cdot\mathbf{u} = \frac{3+1}{\sqrt{2}} = 2\sqrt{2} \approx 2.83$ (signed shadow length); projection vector $= 2\sqrt{2}\cdot\frac{1}{\sqrt{2}}(1,1) = (2,2)$.
4. $\dfrac{\mathbf{a}\cdot\mathbf{v}}{\mathbf{v}\cdot\mathbf{v}}\mathbf{v} = \dfrac{4}{2}(1,1) = (2,2)$: the *same* projection vector. The projection does not depend on how long the direction vector is — only the coordinate does.

Definition — basis

A **basis**: reference axes in which every vector is written in exactly one way. **Orthonormal**: the axes are perpendicular and of length one — and then every coordinate is a projection, $x_i = \mathbf{x} \cdot \mathbf{e}_i$.



The pink arrow never moves. Grey axes: $(3, 1)$. Tilted axes: $(2\sqrt{2}, -\sqrt{2})$.

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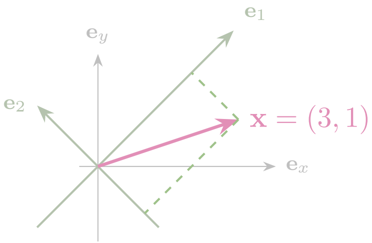

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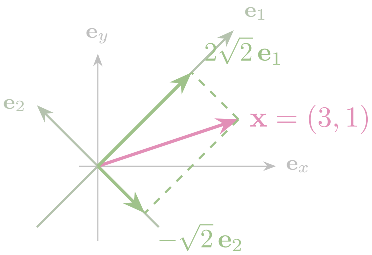

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Change of basis: out with $\mathbf{B}^\top$, back with $\mathbf{B}$

$$\mathbf{x}_{\text{new}} = \mathbf{B}^\top\mathbf{x}$$

measure: row i of $\mathbf{B}^\top$ is $\mathbf{e}_i$, $x_i = \mathbf{e}_i \cdot \mathbf{x}$

$$\begin{tikzcd} \mathbf{x}\ \text{(old axes)} \arrow[r, bend left] & \mathbf{x}_{\text{new}}\ \text{(new axes)} \arrow[l, bend left] \end{tikzcd}$$

$$\mathbf{x} = \mathbf{B}\,\mathbf{x}_{\text{new}}$$

rebuild: column i of $\mathbf{B}$ is $\mathbf{e}_i$, $\mathbf{x} = \sum_i x_i\mathbf{e}_i$

Useful fact

$\mathbf{B} = (\mathbf{e}_1\ \mathbf{e}_2)$ holds the new Basis vectors as *columns*. Orthonormality is exactly $\mathbf{B}^\top\mathbf{B} = \mathbf{I}$, i.e. $\mathbf{B}^{-1} = \mathbf{B}^\top$: no inversion is ever needed.

Both arrows are matrix–vector products, so both are linear combinations: what differs is *which reading means something*. Out, read $\mathbf{B}^\top$ by rows (one dot product per axis); back, read $\mathbf{B}$ by columns (one basis vector per coefficient).

Two checks: rebuild, and compare lengths

Useful fact

1. **Rebuild.** Recompute $x_1\mathbf{e}_1 + x_2\mathbf{e}_2$. If it returns $\mathbf{x}$, the coordinates are right: *this check is decisive*, because the writing of a vector in a basis is unique.
2. **Compare lengths.** $x_1^2 + x_2^2 = \|\mathbf{x}\|^2$. Faster — but partial: it cannot see a swap or a sign error.

The second is Pythagoras — and it is exactly what fails for non-perpendicular axes.

Example

Axes rotated by $45°$: $\mathbf{e}_1 = \frac{1}{\sqrt{2}}(1,1)$, $\mathbf{e}_2 = \frac{1}{\sqrt{2}}(-1,1)$. Express $\mathbf{x} = (3,1)$ in this new basis.

$$x_1 = \mathbf{x}\cdot\mathbf{e}_1 = \frac{1}{\sqrt{2}}(3+1) = 2\sqrt{2}, \qquad x_2 = \mathbf{x}\cdot\mathbf{e}_2 = \frac{1}{\sqrt{2}}(-3+1) = -\sqrt{2}.$$

Rebuild — the way back, from $\mathbf{x}_{\text{new}} = (2\sqrt{2}, -\sqrt{2})$

$$2\sqrt{2}\,\mathbf{e}_1 - \sqrt{2}\,\mathbf{e}_2 = (2,2)+(1,-1) = (3,1) \checkmark$$

Lengths

$$x_1^2 + x_2^2 = 8 + 2 = 10 = \|\mathbf{x}\|^2 \checkmark$$

Rebuilding *is* the reverse change of basis: it takes the coordinates $(2\sqrt{2}, -\sqrt{2})$ and expresses the same arrow back in the canonical basis $(1,0), (0,1)$, where it reads $(3,1)$. And $\mathbf{B}^\top$ replaces $\mathbf{B}^{-1}$ here **only because this basis is orthonormal** — with skewed axes you must genuinely invert.

Two typical errors, and the check that detects each

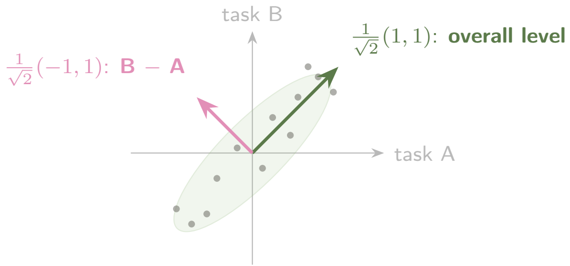
Watch out

$$(3\sqrt{2},\ -2\sqrt{2}) \qquad \text{and} \qquad \sqrt{2}\,(\mathbf{e}_1 - 2\mathbf{e}_2)$$

First: the projections were never taken — the **length** check kills it ($18 + 8 = 26 \neq 10$).
Second: the two coefficients were swapped — lengths still give 10, so only **rebuilding** catches it ($(3, -1) \neq (3, 1)$).

One error per check: that is why you run both.

Choosing axes aligned with the variability of the data



- **Align the axes with the way the data actually varies**

  Each dot is a participant, scored on two similar tasks (centred). The cloud is stretched one way and thin across it: the two raw coordinates say almost the same thing twice. Rotate onto the long and short directions and the new coordinates become "how good overall" and "which task is easier for *this* person" — the covariance $\begin{pmatrix} v & c \\ c & v \end{pmatrix}$ is now diagonal, i.e. the two new scores are uncorrelated. *(The two tasks have the same variance v here, which is what puts the good axes at exactly $45°$.)*

Practice — try it

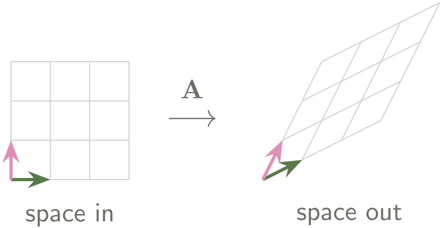
1. Show that $\{\frac{1}{\sqrt{2}}(1,1),\ \frac{1}{\sqrt{2}}(1,-1)\}$ is orthonormal.
2. Write $\mathbf{x} = (2,0)$ in the basis $\mathbf{e}_1 = \frac{1}{\sqrt{2}}(1,1)$, $\mathbf{e}_2 = \frac{1}{\sqrt{2}}(-1,1)$, then rebuild it as a check.
3. Give an orthonormal basis whose first axis points along $(3,4)$.

A matrix as a transformation of space

Definition — matrix–vector product

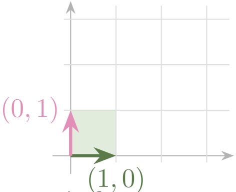
$\mathbf{A}$ of size $m \times n$ sends $\mathbf{x} \in \mathbb{R}^n$ to $\mathbf{A}\mathbf{x} \in \mathbb{R}^m$; output slot i is (row i) $\cdot$ $\mathbf{x}$:

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{21}x_1 + a_{22}x_2 \end{pmatrix}.$$

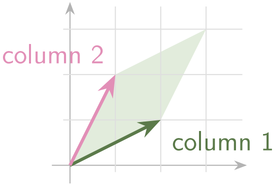


Rotate, stretch, shear or squash — origin fixed, straight lines straight, grid lines evenly spaced.

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \mathbf{A}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad \mathbf{A}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$



before: area 1



after: area $|\det \mathbf{A}| = 3$

- **Why $\det$ is an area**

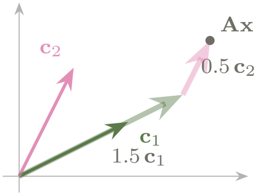
  The unit square is spanned by $(1,0)$ and $(0,1)$; $\mathbf{A}$ sends it to the parallelogram on the two *columns*. The grid is made of copies of that square, so **every** region is stretched by the same factor. That single factor is $|\det \mathbf{A}|$; its sign says whether the plane was flipped.

This same $\mathbf{A}$ returns in Session 2.

- **Intuition**

$$\mathbf{A}\mathbf{x} = x_1\mathbf{c}_1 + x_2\mathbf{c}_2$$

The output is a linear combination of the *columns*.

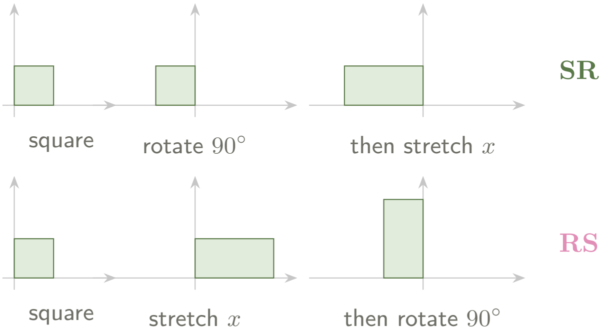


$\mathbf{x} = (1.5,\ 0.5)$

$\Rightarrow$ the set of possible outputs is exactly span(columns).

Rows to compute, columns to understand — and that is what makes *rank* obvious next session.

Order matters: $\mathbf{AB} \neq \mathbf{BA}$



Same two operations, opposite order, different shape. In $\mathbf{AB}$ the *right-hand* matrix acts first; $(\mathbf{AB})^\top = \mathbf{B}^\top\mathbf{A}^\top$ reverses too.

Matrices — worked example

Example

Recode two task scores as *mean* and *difference*: $\mathbf{M} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ -1 & 1 \end{pmatrix}$, scores $\mathbf{s} = (8, 4)$.

$$\begin{tikzcd} \boxed{\begin{array}{l} \text{task A} = 8 \\ \text{task B} = 4 \end{array}} \arrow[d, "\mathbf{M}"] \\ \boxed{\begin{array}{c} \text{mean} = 6 \\ \text{B} - \text{A} = -4 \end{array}} \end{tikzcd}$$

By rows

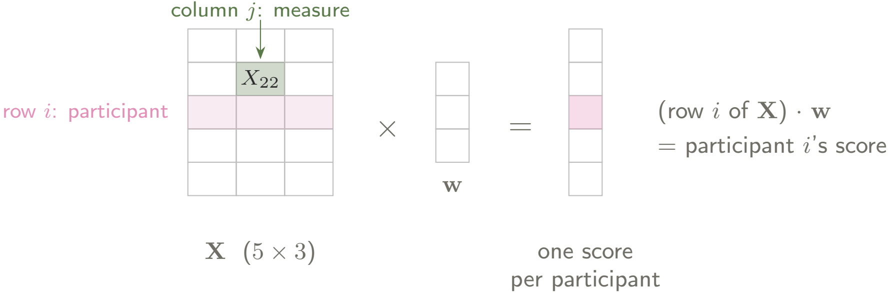
$$\mathbf{Ms} = \begin{pmatrix} (\frac{1}{2}, \frac{1}{2}) \cdot (8, 4) \\ (-1, 1) \cdot (8, 4) \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \end{pmatrix}$$

By columns

$$8\begin{pmatrix} \frac{1}{2} \\ -1 \end{pmatrix} + 4\begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \end{pmatrix} \checkmark$$

Same participant, two numbers in, two numbers out — and the second pair is the one you would actually analyse.

The data matrix



- **One product scores everybody at once**

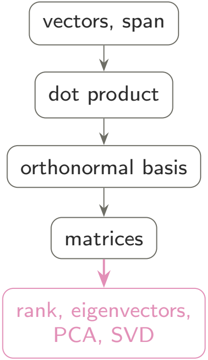
  X_{ij}: **participant** i, **measure** j — rows are people, columns are variables. Then $\mathbf{Xw}$ applies one scoring rule to *everybody at once*, in a single product. Predicting an outcome from several measures is exactly this: the linear part of a regression is $\mathbf{Xw}$.

Practice — try it

1. Compute $\left(\begin{smallmatrix} 2 & 0 \\ 0 & 3 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$; say in words what it does, and what it does to areas.
2. Compute $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} 5 \\ -2 \end{smallmatrix}\right)$. Which geometric operation is it?
3. Compute $\left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & 0 \\ 3 & 1 \end{smallmatrix}\right)$ and the same product the other way. Comment.

Session 1 — summary

- **Span** = what a set of vectors can build; **independence** = no redundancy.
- **Dot product** ⇒ length, angle, and the **projection** picture.
- **Change of basis**: out with $\mathbf{B}^\top$, back with $\mathbf{B}$ — checked twice.
- **Matrix** = transformation: rows to compute, columns to understand.



Next session: what a transformation does to space — and how many dimensions a data table really has.